



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO JAMBAI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

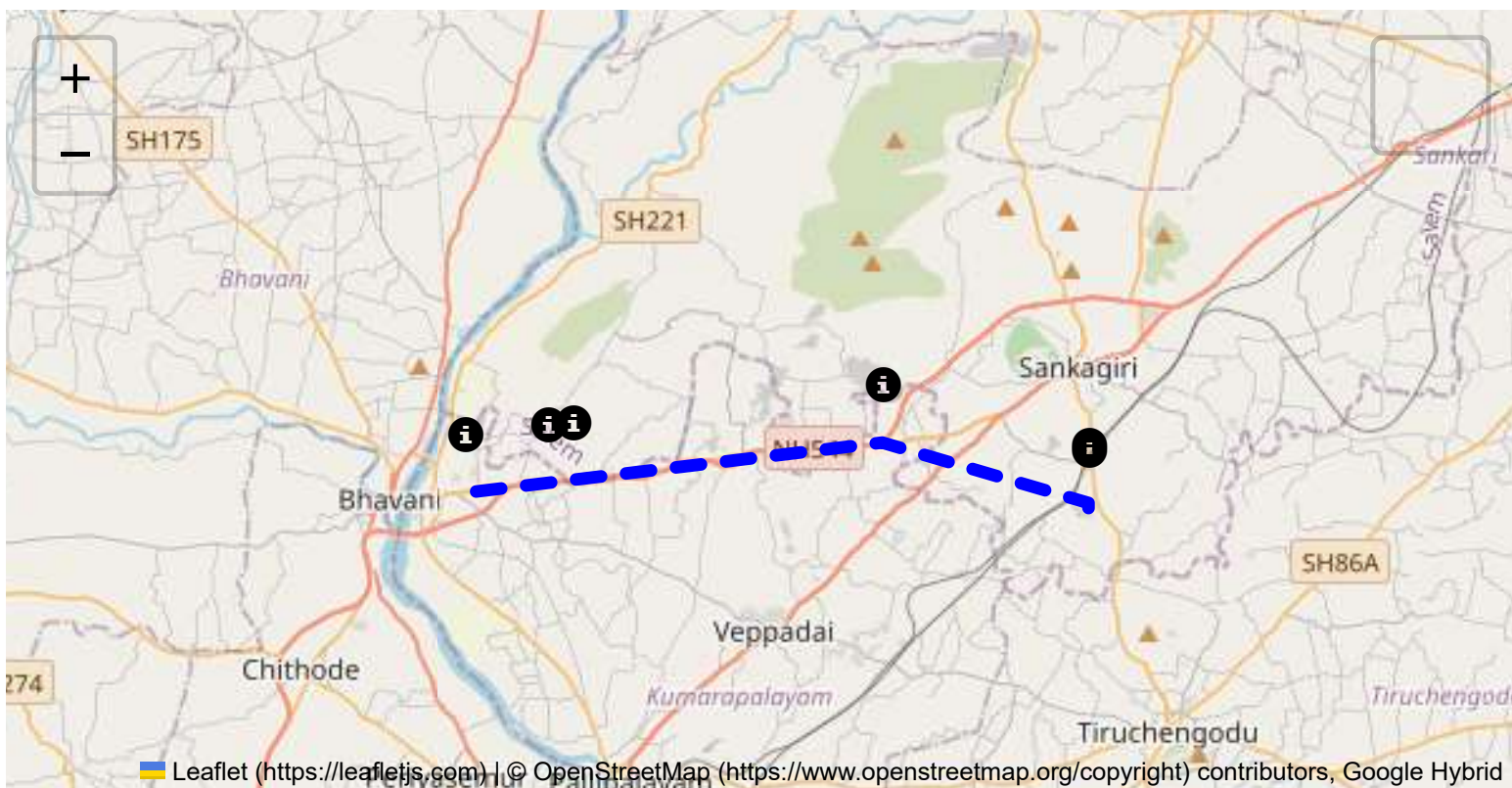
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

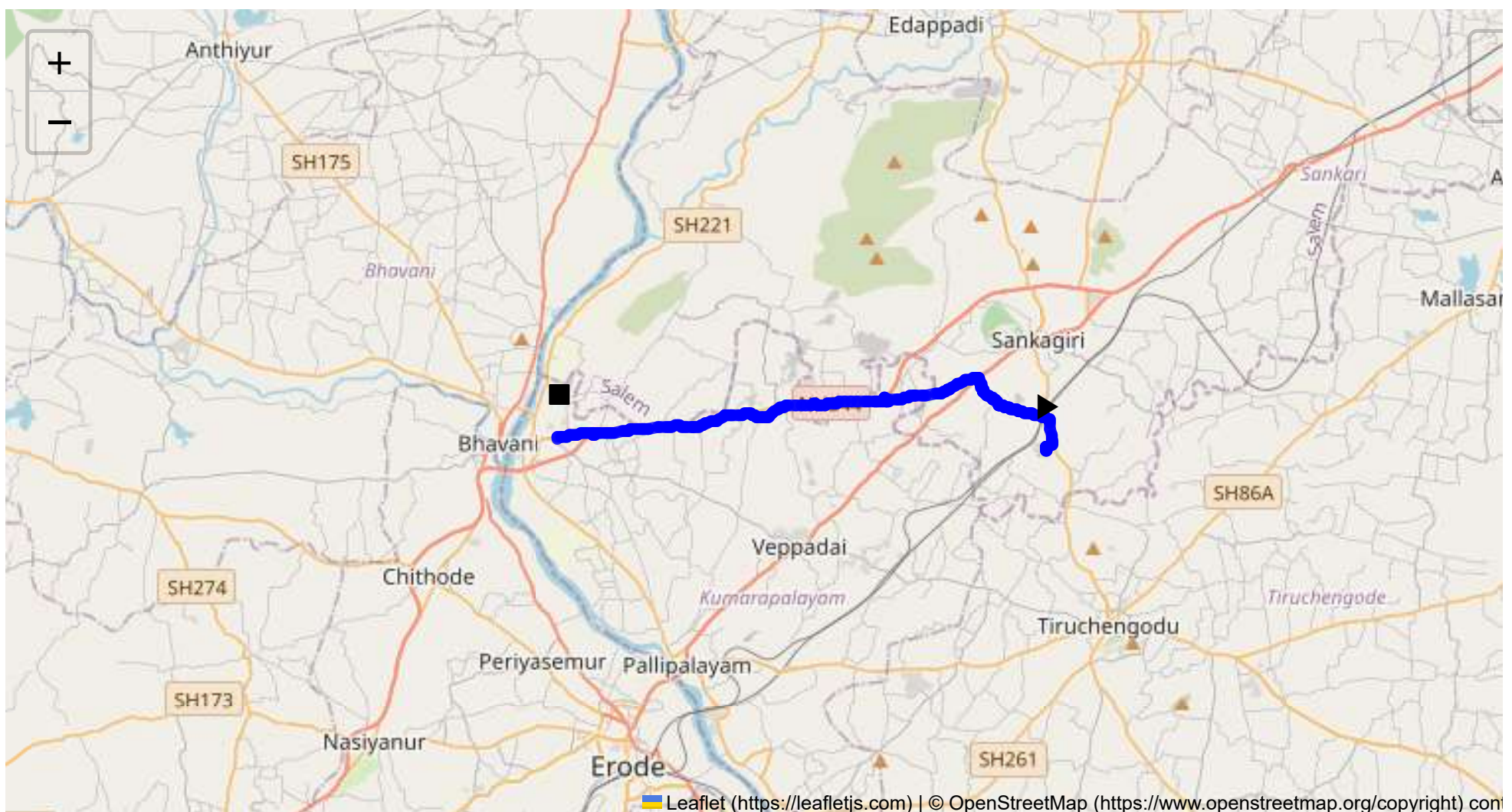
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 21.51 km
Estimated Duration: 0.5 hours
Adjusted Duration (Heavy Vehicle): 0.6 hours
Start: (11.4381, 77.8734)
End: (11.442492, 77.703138)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route extends approximately 21.51 kilometers from Sangagiri to Kammavarpalayam in Tamil Nadu, India. It passes through Puthur, Padaiveedu, Tiruchengode, utilizing major roads such as Bhavani Main Road and Service Road. This route connects prominent rural and semi-urban regions with varied road conditions.

2. Typical Weather Conditions and Potential Weather-related Hazards

Tamil Nadu experiences a generally tropical climate with hot summers, rainy monsoons, and mild winters. During the monsoon season (October to December), heavy rainfall can result in reduced visibility and slippery roads. The summer (March to May) can introduce challenges of high temperatures, potentially impacting vehicle performance.

3. Analysis of Traffic Patterns

- **Peak Hours:** The route typically experiences peak traffic between 8 AM - 10 AM and 5 PM - 7 PM on weekdays.
- **Congestion-Prone Areas:** The town centers of Tiruchengode and Kammavarpalayam are known for congestion, especially near marketplaces and schools.

4. Assessment of Road Quality and Infrastructure

The road quality varies along the route:

- **Sangagiri to Tiruchengode:** Generally maintained, with occasional potholes and narrow lanes in Puthur.
- **Tiruchengode to Kammavarpalayam:** Better quality roads but watch for ongoing construction projects that can cause sudden detours and roadblocks.

5. Suggestions for Alternative Routes for Emergencies

In emergencies, an alternative route via State Highway 79, bypassing Puthur, may offer a quicker and less congested path, although it is a longer detour.

6. Summary of Local Regulations Affecting Hazardous Material Transport

The transportation of hazardous materials is regulated under the Motor Vehicles Act with requirements for clear labeling and documentation. Restrictions may apply during peak hours in densely populated areas. Drivers must ensure appropriate adherence to local material transport laws.

7. Overview of Historical Incidents

There have been occasional incidents involving heavy vehicles on this route linked to monsoon-related poor road conditions. No major hazardous material incidents have been reported in recent history.

8. Environmental Considerations and Sensitive Areas

The route passes through agricultural land, requiring careful handling of hazardous materials to prevent environmental contamination. Areas near Täpo on Bhavani Main Road warrant particular caution due to nearby water bodies.

9. Analysis of Communication Coverage

Mobile network coverage is generally good across the route, with potential weak areas near rural localities between Puthur and Padaiveedu. GPS functioning may be sporadic in heavily congested urban zones.

10. Estimated Emergency Response Times

- Sangagiri to Puthur:** Approximately 20-30 minutes for emergency services.
- Puthur to Tiruchengode:** 15-25 minutes, due to more accessible location.
- Tiruchengode to Komarapalayam:** 25-35 minutes, dependent on traffic congestion.

11. Overall Summary of Risk Assessment

This route poses moderate risks primarily due to weather conditions and urban congestion in Tiruchengode, compounded by periodic road infrastructure challenges. Adequate precautions, timely monitoring of weather, and opting for alternative routes during peak hours, can enhance safety for hazardous material transport. Maintaining strong communication links and adhering to transport regulations are crucial to mitigate risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43961, 77.87341	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44898, 77.87410	30 KM/Hr
5	Turn	Medium	11.45348, 77.85706	30 KM/Hr
6	Turn	Medium	11.46318, 77.84913	30 KM/Hr
7	Turn	Medium	11.45697, 77.82719	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
10	Turn	High	11.45636, 77.81678	15 KM/Hr
11	Blind Spot	Blind Spot	11.45543, 77.81740	10 KM/Hr
12	Turn	High	11.45516, 77.81346	15 KM/Hr
13	Turn	High	11.45521, 77.81343	15 KM/Hr
14	Blind Spot	Blind Spot	11.44530, 77.72783	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Blind Spot	Blind Spot	11.44513, 77.72786	10 KM/Hr
16	Turn	High	11.44377, 77.71562	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital	11.4525596, 77.7749426	30 km/h	Medium
4	hospital	Dhanvantri Multi Speciality Hospital	11.451362, 77.766602	30 km/h	Medium
5	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	30 km/h	Medium
6	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Vivekanandha Engineering College	11.4589312, 77.7899284	30 km/h	Medium
2	marketplace	Monday market	11.452863, 77.775989	30 km/h	Medium
7	school	JKK Matriculation and Higher Secondary school	11.4446843, 77.7066207	30 km/h	Medium
8	school	JKK Girls Higher Secondary school	11.4437747, 77.7066298	30 km/h	Medium
9	college	Government College of Education	11.4476704, 77.705553	30 km/h	Medium
10	school	Government Girls Hr Sec School	11.4456929, 77.7007497	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45348, 77.85706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45697, 77.82719



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45636, 77.81678



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45516, 77.81346



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45521, 77.81343



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44530, 77.72783



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44513, 77.72786



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44377, 77.71562
